

Basis

Minimum vectors jo poori space ko span kar sakein

01 Opening Hook

STORY TIME

Socho tumhare paas ek address system hai jisme sirf 2 directions se koi bhi jagah describe ho sakti hai - 'North-South' aur 'East-West'. Tumhe teesri direction (jaise 'North-East') ki zaroorat nahi hai, kyunki woh pehle dono se hi ban sakti hai. Yeh 'minimum zaroori directions' ka set hai ****Basis****.

02 Why Does This Concept Exist?

THE PROBLEM

Span batata hai ki vectors se kya-kya banaya ja sakta hai, lekin agar humare paas zaroorat se zyada vectors hain (redundant), toh computation waste hoti hai. Basis invent hua taaki hum 'sabse chhota, sabse efficient' set of vectors dhoond sakein jo poori space ko describe kar sake, bina kisi redundancy ke.

03 Explain Like I'm 10

SIMPLE WORDS

Socho tumhare paas ek paint set hai. Agar tumhare paas Red, Blue, aur Purple (jo Red+Blue se banta hai) teeno hain, Purple 'zaroori' nahi hai - woh already Red aur Blue se ban sakta hai. Basis matlab hai sirf zaroori colors rakhna - Red aur Blue - jisse tum baaki saare colors bana sako, bina extra/redundant colors ke.

School coordinate system mein bhi x-axis aur y-axis 'basis vectors' hain - unse tum poore 2D plane ka koi bhi point describe kar sakte ho.

04 Build Intuition First (No Maths Yet)

FEEL IT FIRST

Basis vectors do properties satisfy karte hain: (1) woh poori space ko span karte hain, (2) woh linearly independent hain (koi bhi redundant nahi hai). Ek n -dimensional space ka basis hamesha exactly n vectors ka hota hai.

05 Ab Thoda Maths

FORMULA TIME

Standard Basis ($\mathbb{R}^2$):

$e_1 = [1, 0]$ aur $e_2 = [0, 1]$ - yeh standard basis hain. Koi bhi 2D vector $[x, y]$ inko combine karke likha ja sakta hai: $x \cdot e_1 + y \cdot e_2$.

Basis Change:

Ek hi vector ko different basis mein alag tarique se represent kiya ja sakta hai - jaise PCA naya basis (principal components) use karta hai data represent karne ke liye.

06 Visualize Everything

PICTURE IT

Standard Basis in $\mathbb{R}^2$:

y
|
e2=[0,1]
|
|_____ e1=[1,0]_____ x

Koi bhi vector $[3, 4] = 3 \cdot e_1 + 4 \cdot e_2$

07 AI Connection

WHERE AI USES THIS

PCA's New Basis: PCA data ke liye ek naya basis (principal components) dhoondta hai jisme data ki variance sabse zyada 'axis-aligned' ho.

Fourier/Frequency Basis: Audio/signal processing mein Fourier basis use hota hai signals ko frequency components mein represent karne ke liye.

08 Real Industry Example

REAL COMPANIES

Google DeepMind (Research): Neural network interpretability research mein 'basis directions' dhoondi jaati hain jo specific concepts (jaise 'curve detector' neurons) represent karti hain.

Signal Processing companies: Audio compression algorithms (jaise MP3) frequency-basis use karte hain data ko efficiently represent karne ke liye.

09 Explain Like I'm Going To Build It

HOW YOU'D CODE IT

Practically, tum jab `nn.Linear` layer banate ho, uske weight rows ek naya 'basis' seekhte hain jisme input features ko re-represent kiya jaata hai. PCA implementation mein `pca.components_` woh naya basis hai jo scikit-learn deta hai.

10 Common Misconceptions

BEGINNER MISTAKES

Galat soch: Basis hamesha unique hota hai.

Sahi baat: Ek hi vector space ke multiple valid basis ho sakte hain – jaise standard basis $[1,0], [0,1]$ ke alawa rotated basis bhi valid hai, jab tak woh span + independence conditions satisfy kare.

Galat soch: Zyada vectors better basis banate hain.

Sahi baat: Basis mein exactly utne hi vectors hone chahiye jitni space ki dimensionality hai – extra vectors redundant honge, kam vectors poori space cover nahi karenge.

11 Interview Questions

INTERVIEW CORNER

[Beginner] Q: Basis kya hota hai?

A: Minimum, linearly independent vectors ka set jo poori vector space ko span kar sake.

[Intermediate] Q: Standard basis aur custom basis (jaise PCA se) mein kya farak hai?

A: Standard basis simple axes hote hain (jaise x, y axis); custom basis (PCA) data ke structure ke hisaab se choose kiya jaata hai taaki representation zyada meaningful/efficient ho.

[Advanced] Q: Basis change (change of basis) practically AI mein kahan use hota hai?

A: PCA, whitening transformations, aur kuch attention mechanisms mein data ko naye basis mein project kiya jaata hai taaki important patterns zyada clearly separate ho sakein.

12 Exercises

PRACTICE TIME

[Easy] $\mathbb{R}^2$ ka standard basis kya hai?

Solution: $[1,0]$ aur $[0,1]$.

[Medium] Kya $\{[1,0], [0,1], [1,1]\}$ ek valid basis hai $\mathbb{R}^2$ ke liye?

Solution: Nahi – yeh 3 vectors hain jabki $\mathbb{R}^2$ ka basis sirf 2 chahiye; $[1,1]$ redundant hai kyunki woh $[1,0]$ aur $[0,1]$ ka combination hai.

[Hard] PCA ka naya basis standard basis se kaise better hota hai data ke liye?

Solution: PCA ka basis data ki actual variance directions ke saath align hota hai, jisse fewer dimensions mein zyada information capture ho sakti hai, jabki standard basis arbitrary directions use karta hai jo data ke pattern se match nahi karta.

13 Golden Summary

Concept	Easy Meaning
Basis	Minimum, independent vectors jo poori space span karein
Standard Basis ($\mathbb{R}^2$)	$[1,0]$ aur $[0,1]$
Basis Size	Hamesha space ki dimensionality ke barabar
Change of Basis	Data ko naye basis mein re-represent karna (jaise PCA)
Non-uniqueness	Ek space ke multiple valid basis ho sakte hain

14 One-Line Memory Trick

“Basis woh sabse chhota 'toolkit' hai jisse poori space ban sakti hai.”